

# Damien MAÎTRE

MSc Engineering Student — CFD, HPC & Numerical Methods

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## Education

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| September 2026<br>February 2027 | <b>MSc Mobility &amp; Physics engineering</b><br><i>Student exchange at Politecnico di Milano, Italy</i><br>International economy, transport operation, high performance computing, infrastructure finance                                                                                                                                                                                                                                                                                                                                                                                                                |
| September 2023<br>June 2027     | <b>MSc Engineering — Fluid dynamics, Structural mechanics, Prototyping</b><br><i>École nationale des ponts et chaussées (ENPC), Paris</i><br>Structural mechanics and large-structure design, FEM applied to structural analysis, composite materials, metallurgy, complete fluid dynamics module, HPC, combustion <ul style="list-style-type: none"><li>• <b>Tools:</b> OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, gmsh</li><li>• <b>Projects:</b> Underwater sea-glider simulation and prototype ; closed-wing aircraft design and optimization ; wooden suspension bridge design, verification and construction</li></ul> |
| September 2021<br>June 2023     | <b>Bachelor — Classes préparatoires</b><br><i>Lycée Kléber, Strasbourg</i><br>Intensive programme in mathematics, physics and engineering sciences                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Research Projects

### Complete CFD developer-first suite

*Personal project — OpenFOAM, Gmsh (forked), Python, C++*

Open and user-friendly aerodynamical loads studies, tailored for very high  $Re$  axisymmetric problems.

### Structural FEM & Eurocode Verification suite

*Personal project — OpenSeesPy, Python*

End to end structure design, visualization and verification tools.

## Professional Experience

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| February 2026<br>July 2026 | <b>Rail infrastructure engineering internship — TRACTEBEL FRANCE</b><br><i>CDG Express project, Saint-Ouen</i> <ul style="list-style-type: none"><li>• Construction compliance verification for the CDG Express, a 2.5B€ high-speed rail project</li><li>• Structural analysis and Eurocode verifications of civil engineering structures</li><li>• Review of structural calculation reports and as-built site surveys</li><li>• Drafting technical reports and recommendations for the client</li></ul> |
| July 2025<br>January 2026  | <b>Digital Twin PM intern — SNCF Voyageurs</b><br><i>Industry &amp; digital solutions, Saint-Denis</i> <ul style="list-style-type: none"><li>• Management of a 150k€ digital project within the rolling stock maintenance industry</li><li>• Coordination of 15 3D artists across the country's maintenance plants</li><li>• Project ownership and reporting to internal clients (plant directors, Industrial Director, etc.)</li><li>• End-to-end procurement and tendering process</li></ul>           |

## Technical Skills

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|----------------------|----------------------------------------------------------------|
| <b>CFD &amp; FEM</b> | OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, Gmsh                   |
| <b>Programming</b>   | 10+ years of experience with Python, C++ and UNIX environments |

## Languages

**French** — Native · **English** — Proficient (C1) · **Italian** — Working proficiency (B2) · **German** — Working proficiency (B2)

## Activities & Community Involvement

### Rescuer — French Red Cross

*Volunteer first-aid responder*

First-aid responder at public events (marathons, festivals, Paris-Plage) and on-call with Paris Fire Brigade

### Director — Student Musical Ensemble

*BridgeWay Comédie Musicale, Champs-Sur-Marne*

Directed an ensemble of 75 artists and a creative/technical leadership team of 12

### General Secretary — Student Association

*Union des Elèves de l'Ecole Nationale des Ponts et Chaussées*

Management of a team of 15 for professional and social events with 1,000+ attendees, Administrative management of a registered French non-profit (Loi 1901).